DEEPP FINAL PROJECT | BANKING INDUSTRY

Banking Credit Risk Analysis Berka Retail Bank

How Berka can cut its problematic loan rate from 11.1% to below 5% with data it already holds

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Executive Summary

Two pre-loan reviews can cut problematic loans from 11.1% to 5%, using data Berka already holds

11.1%
Problematic loan rate today (76 of 682 loans)

< 5%
Target rate within 12 months

1.05M
Transactions analysed with Python, DuckDB and Tableau

5.0%
Rate after the two pre-loan reviews

Three key findings

1. **Overdraft before the loan is the strongest signal**
   All 27 accounts that went below zero before the loan went bad

2. **Heavy instalments drive bad loans, not big loans**
   The heaviest instalment group fails at 19.3%, the lightest at 5.8%

3. **Demographics barely matter**
   Gender shows almost no difference in risk (1.1x)

RECOMMENDATION

Launch the overdraft review now, add a review of the top 20% risk scores, and keep an instalment cap as a buffer.

11.1% → 5.0%

Project Background

Berka earns from lending, yet 1 in 9 loans ends up problematic

THE BANK

Berka Retail Bank serves individual customers (B2C). It earns mainly from loan interest, plus card and transaction fees.

THE DATA

Public Czech Berka dataset (PKDD'99), 1993 to 1998. Every account behaviour is recorded before a loan is approved.

1 in 9

loans end up problematic

11.1%, or 76 of 682 loans

5,369
clients

4,500
accounts

1.05M
transactions

682
loans

Project Goals

One clear problem, one decision owner, and a scope the data can answer

PROBLEM STATEMENT

How can the credit team reduce Berka's problematic loan rate from 11% to below 5% within 12 months by identifying risk patterns in pre-loan transaction and account data?

Who is involved (DARCI)

D	**Decider**	Head of Credit Risk
A	**Accountable**	Credit Risk Manager
R	**Responsible**	Data Analyst (me)
C	**Consulted**	Loan Officers, Collections Team
I	**Informed**	Senior Management, Branch Heads

Scope

IN SCOPE

Pre-loan transactions, account and loan activity, and good (A, C) vs problematic (B, D) loan outcomes.

OUT OF SCOPE

Revenue and profit, customer satisfaction, and marketing channels. None of these are in the data.

Root Cause Analysis

Four measurable ways a borrower can slip into trouble

Problematic loans

11.1%

76 of 682 loans

1. **Weak account liquidity**
   The savings cushion is thin, the balance often close to zero
2. **Cash flow strain**
   More money goes out than comes in each month
3. **Loan burden**
   The instalment is too heavy for monthly income
4. **Thin banking history**
   Few transactions, so the bank barely knows the customer

The four causes are mutually exclusive and collectively exhaustive. Loan status is the outcome, not a cause.

Hypotheses and Metrics

Each cause becomes one testable hypothesis with one metric

OVERALL KPI

Problematic Loan Rate = (status B + D) ÷ total loans = 11.1%

Root cause	Hypothesis: if … then …	Key metric
Weak account liquidity	If the average balance before the loan is low, then the loan is more likely to go bad	Average pre-loan balance
Cash flow strain	If money out is high relative to money in, then default risk rises	Outflow ÷ inflow
Loan burden	If the monthly instalment is large relative to inflow, then the loan is more likely to fail	Instalment ÷ monthly inflow
Thin banking history	If there are few transactions before the loan, then risk is higher	Count of pre-loan transactions

Methodology

CRISP-DM workflow, focused on decisions rather than code

1

Business understanding

Owner, goal and possible causes

>

2

Data understanding

8 tables, 682 loans

>

3

Data preparation

Clean dates, keep outliers

>

4

Analysis and modelling

Compare groups, build a risk score

>

5

Evaluation and dashboard

Simulate impact, build the dashboard

No-leakage rule

Only transactions before the loan date are used, so every signal is available at approval time.

Outliers kept, not deleted

2,999 negative-balance transactions became the strongest risk signal in the project.

TOOLS

Python

DuckDB

Google Colab

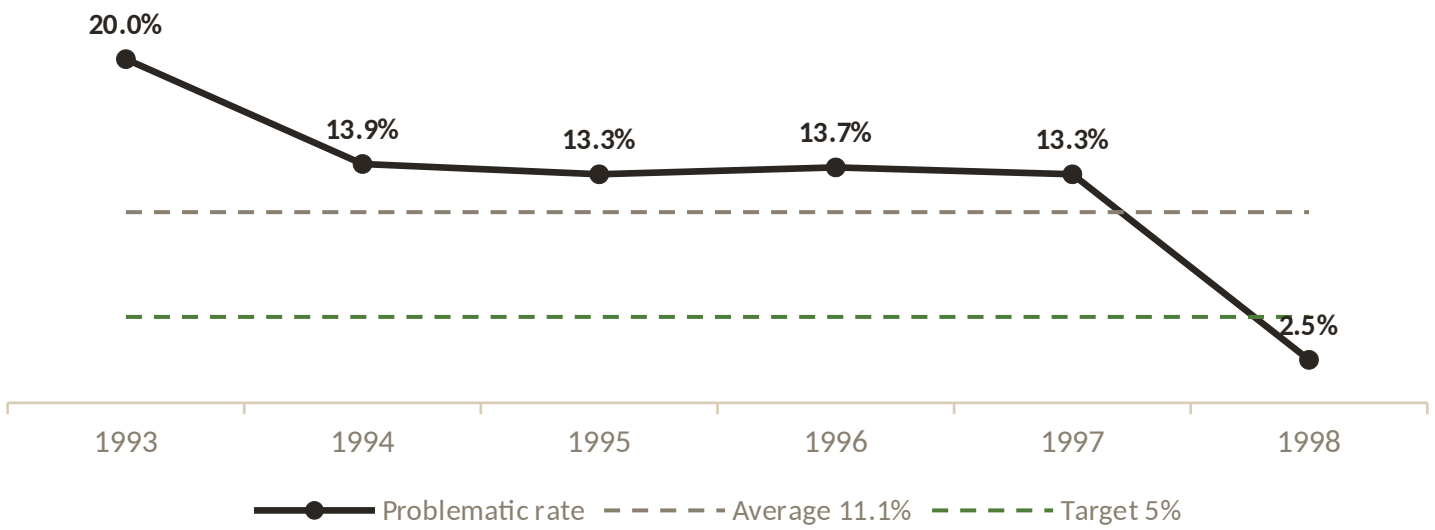
Tableau Public

KPI and Trend

1 in 9 loans goes bad, more than twice the 5% target

682	76	11.1%	15.6M Kč
loans analysed	problematic loans (B + D)	problematic rate	principal at risk

Problematic rate by the year the loan was given (%)



WHAT IT MEANS

The 1998 drop to 2.5% is not real progress. Those loans were too new to go bad. The true rate is 13 to 14%.

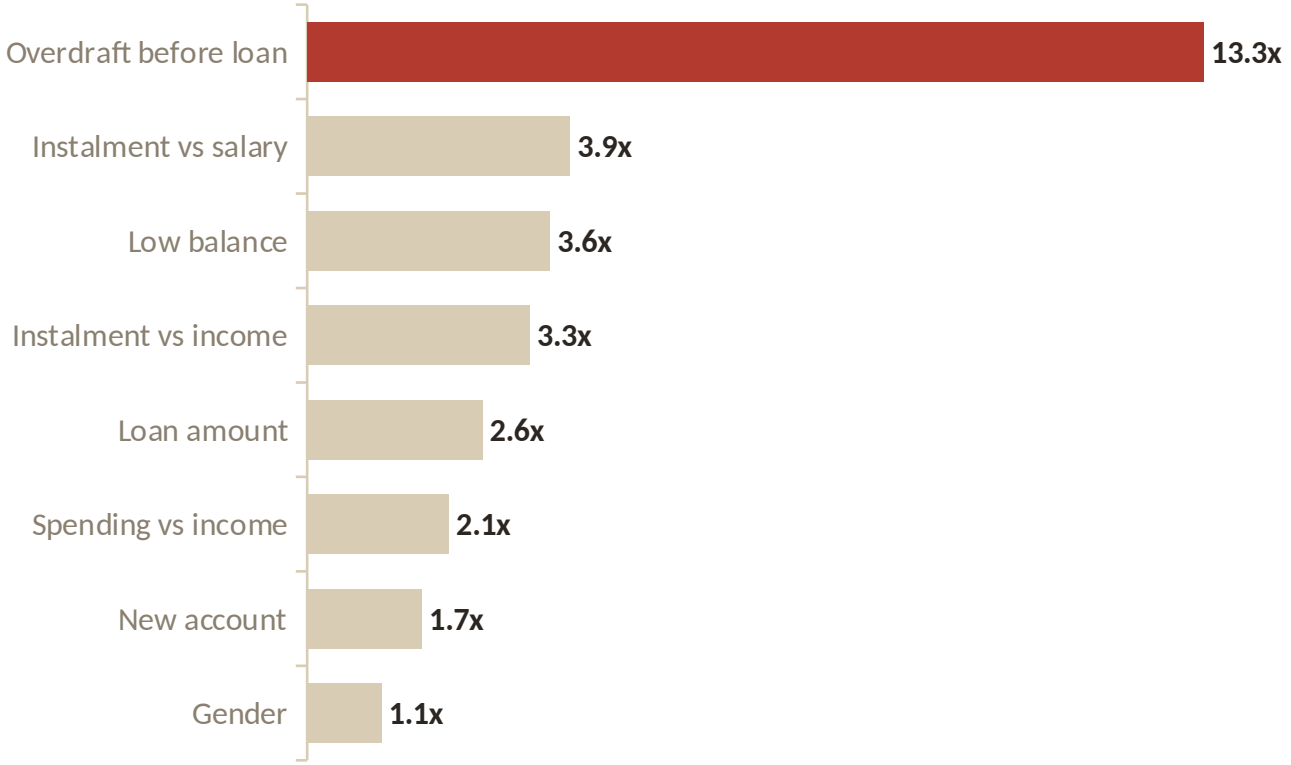
WHAT TO DO

Compare loans of the same age, not one overall number.

Driver Ranking

Account behaviour predicts risk, demographics do not

Risk lift: how many times riskier the worst group is than the safest (1.0x = no difference)



13.3x

riskier when the account went below zero before the loan

WHAT IT MEANS

Risk shows in how the account behaves, not in who the customer is. Gender barely matters (1.1x).

WHAT TO DO

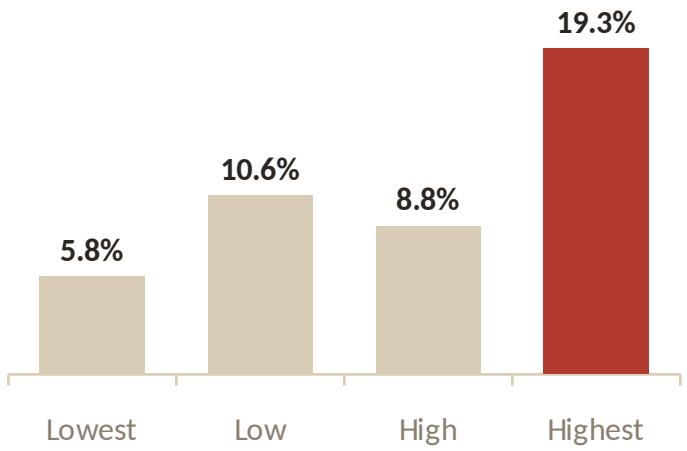
Check account behaviour before approving a loan.

Affordability, Liquidity and Loan Size

The size of the monthly instalment matters more than the size of the loan

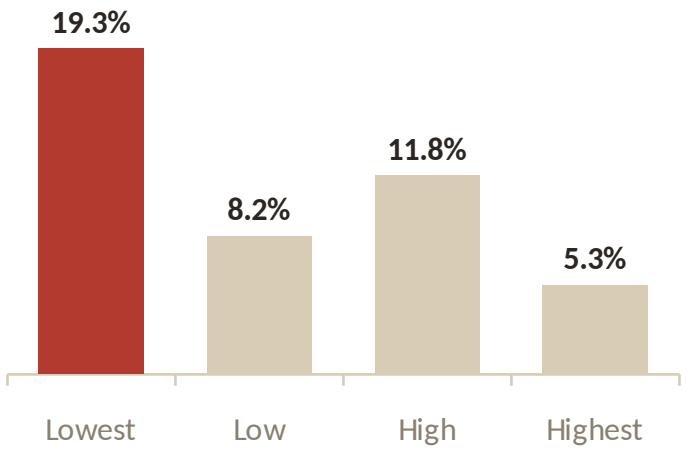
Instalment vs monthly income

Lowest to highest burden



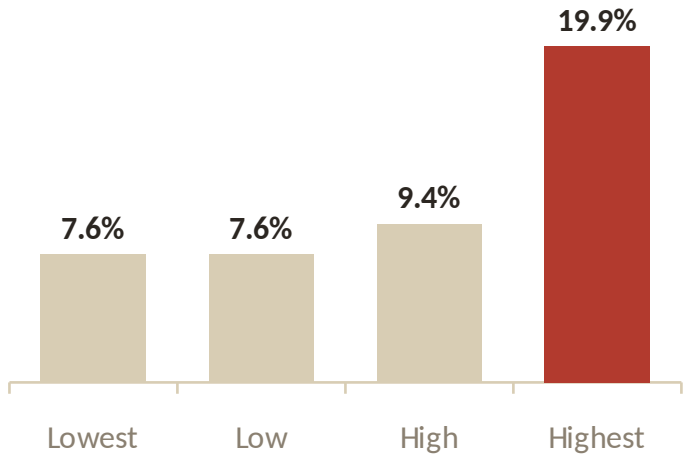
Average account balance

Lowest to highest balance



Loan amount

Smallest to largest loan



WHAT IT MEANS

Heavy instalments and low balances both fail at 19.3%. Loan amount stays flat until the largest group.

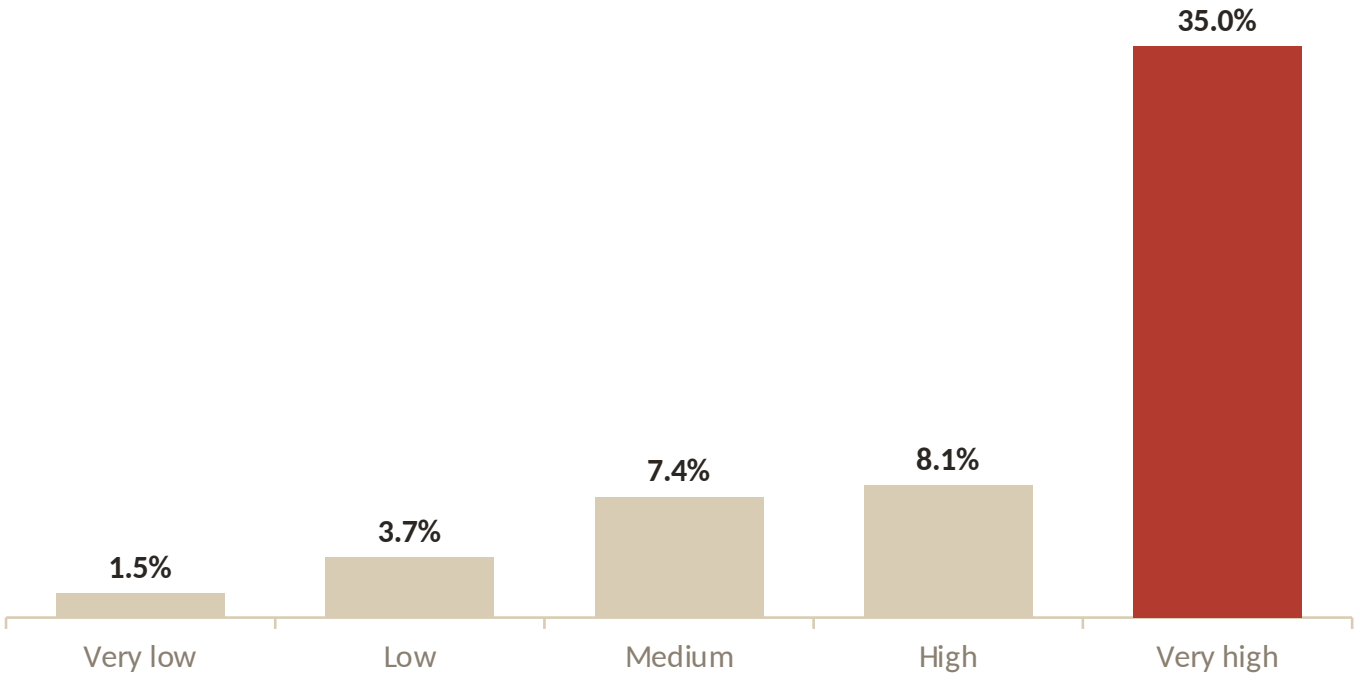
WHAT TO DO

Cap the instalment at 17% of the customer's monthly income.

Risk Score Validation

The riskiest 20% of scores hold 48 of 76 problematic loans

Problematic rate in five equal groups, from lowest to highest risk score



Method: logistic regression, 8 features, 5-fold cross validation

63%

of problematic loans sit in the riskiest 20% of applicants

WHAT IT MEANS

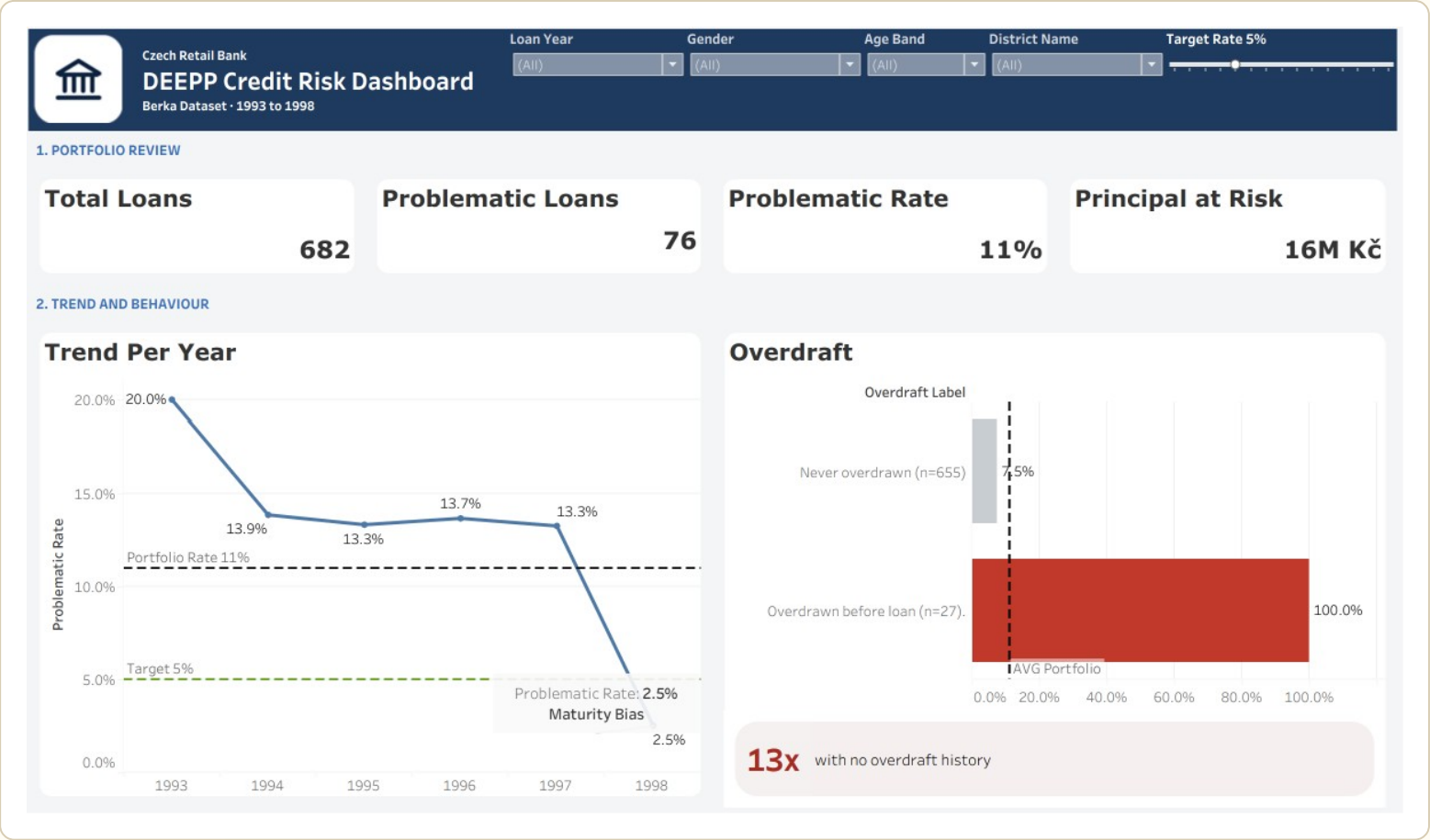
A simple score built from data the bank already has spots most risky applicants.

WHAT TO DO

Send the riskiest 20% to manual review.

Dashboard Overview

One Tableau dashboard shows where risk sits and what fixes it



Red marks the risk group only. Dashed lines show the 11.1% average and the target set by the slider.

BUILT FOR

Head of Credit Risk and loan officers, no code needed

FILTERS

Loan year, gender, age band, district and a Target Rate slider

PANELS

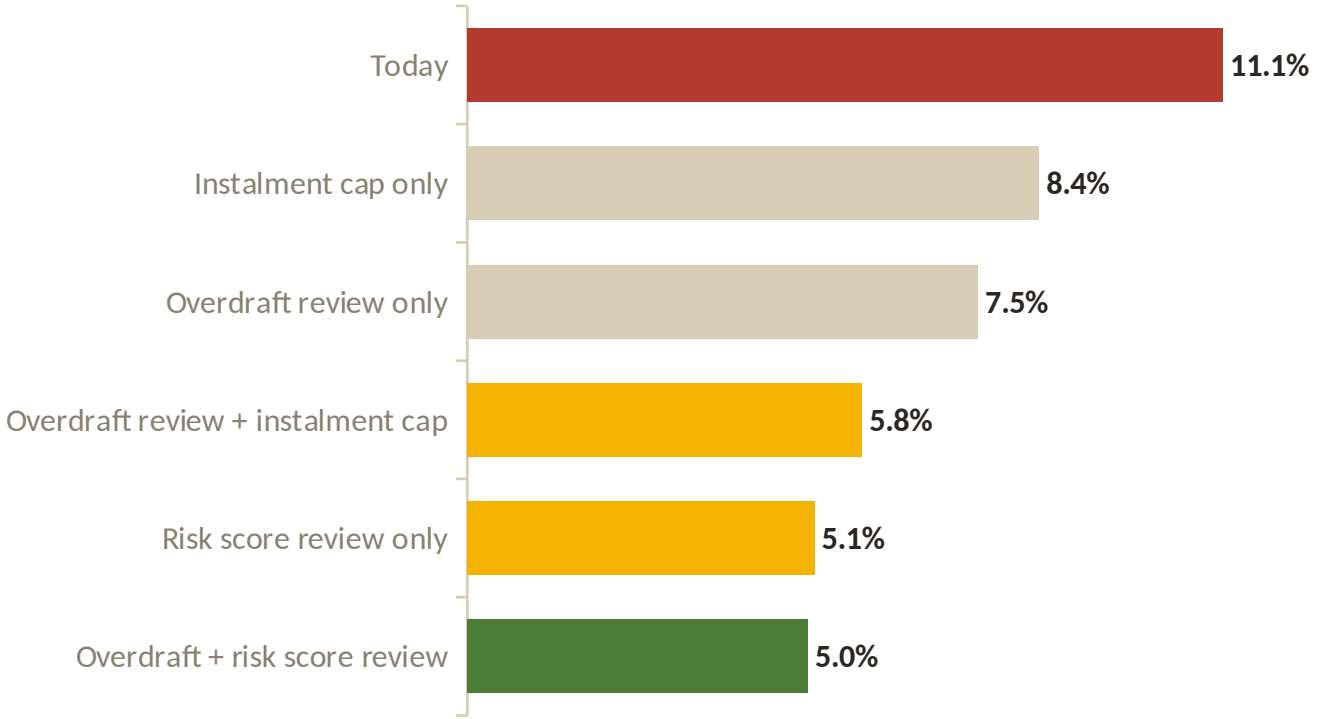
Four sections: portfolio review, trend and behaviour, who is at risk, model score and recommendation

Open dashboard on Tableau Public

Business Impact Simulation

Overdraft review plus risk score review brings the rate to 5%

Problematic rate left after each action (flagged loans declined or restructured)



Green reaches the 5% target · gold is close · grey is a partial fix · red is today

49 of 76

problematic loans removed, checking about 1 in 5 applicants

WHAT IT MEANS

Only the combination of both reviews reaches the 5% target.

TRADE-OFF

About 89 good applicants get a manual check. They are not automatically rejected.

Limitations and Next Steps

Stated openly so the credit team knows how far to trust each number

!

Small event count

Only 76 problematic loans. The overdraft group of 27 accounts needs revalidation.

!

Income is a proxy

No individual income in the data, so district average salary stands in.

!

Association, not causation

A low balance may be a symptom of the same pressure, not its cause.

!

Period-bound data

1993 to 1998 Czech economy. Directions carry over, thresholds need recalibration.

NEXT STEPS

1. Pilot the overdraft and risk score reviews for 3 months
2. Track the rate by loan vintage on the dashboard
3. Revalidate the score on a larger sample before any automatic decline

Recommendations

Four actions, one path from 11.1% to 5%, all using data the bank already has

1

Send every pre-loan overdraft to manual review

Based on: all 27 overdrawn accounts went bad

11.1% → 7.5%, no good loan lost

2

Review the top 20% of risk scores

Based on: this group holds 48 of 76 problematic loans

With action 1: 5.0%, on target

3

Cap the instalment at 17% of monthly income

Based on: the heaviest instalments fail at 19.3%

Keeps a buffer below 5%

4

Compare loans of the same age in reports

Based on: new 1998 loans look falsely safe

Prevents false signs of progress

Appendix A | Dataset and Cleaning

Eight tables linked by account_id, with every cleaning decision documented

Table	Rows
client	5,369
disp	5,369
account	4,500
loan (target)	**682**
trans	1,056,320
order	6,471
card	892
district	77

Area	Issue	Decision
Data types	Dates stored as YYMMDD integers, coded birth_number	Converted with century 19, decoded gender and birth date
Missing values	trans: bank 74%, account 72%, k_symbol 51% blank	Kept every row, imputed nothing. Blanks are structural
Duplicates	None in source tables, the risk comes from joins	Filtered disp to OWNER, asserted 682 loans
Outliers	120,215 amounts flagged by IQR, 2,999 negative balances	Kept all. Negative balances became the overdraft flag

disp is filtered to OWNER so the 682 loans stay 682 after the join. Zero orphan rows across all eight foreign keys.

Appendix B | How to Read the Dashboard

Quartiles split the 682 loans into four equal groups of about 170 loans

Panel	What it shows	Key insight
Trend per year	Rate by loan year vs the 11.1% average and the slider target	1998 at 2.5% is maturity bias, mature years sit at 13 to 14%
Overdraft	Accounts that went below zero before the loan vs the rest	All 27 overdrawn accounts went bad vs 7.5% for the rest (13x)
Affordability	Instalment ÷ own monthly inflow, Q1 lowest burden	Q4 fails at 19.3%, Q1 only at 5.8%
Liquidity	Average pre-loan balance, Q1 lowest balance	Q1 fails at 19.3%, Q4 only at 5.3%
Loan size	Loan amount, Q1 smallest loans	Flat at 7.6 to 9.4% until Q4 jumps to 19.9%
Risk band	Model score split into five equal groups	Very high fails at 35.0% and holds 48 of 76 bad loans
Simulation	Rate left after each action vs the slider target, shaded safe zone	Only overdraft + risk score review reaches 5%

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Thank You

The warning signs are already in the account before the loan is approved.

Dashboard: Tableau Public, Berka credit risk dashboard

Notebook: Google Colab, cleaning and analysis code

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